Write legibly showing all the steps VERY clearly. (Total points = 100).

[1] Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.

(a) For $A, B \in \mathcal{F}$, if $\mu(A \setminus B) = 0$ and $\mu(B \setminus A) = 0$, show that $\mu(A) = \mu(B)$.

(b) Let $A_i \in \mathcal{F}$ be a sequence of (any) events such that $\mu(A_i) = 0$ for all $i \geq 1$. Show that $\mu(\bigcup_{i=1}^{\infty} A_i) = 0$.

(c) Let λ be a finitely additive set function on a σ-algebra $\mathcal{F}$ with $\lambda(\emptyset) = 0$. If $\lambda(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \lambda(A_i)$ for all sequences of disjoint sets in $\mathcal{F}$, prove λ is a measure.

$(10 \times 3 = 30$ points)

(a) We know that $A = (A \setminus B) \cup (A \cap B)$, which is a disjoint union. Thus, $\mu(A) = \mu(A \setminus B) + \mu(A \cap B)$. Since $\mu(A \setminus B) = 0$, we have $\mu(A) = \mu(A \cap B)$. Similarly, $B = (B \setminus A) \cup (A \cap B)$, so $\mu(B) = \mu(B \setminus A) + \mu(A \cap B) = \mu(A \cap B)$. Therefore, $\mu(A) = \mu(B)$.

(b) By countable sub-additivity of measures, we have $\mu(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$. Since $\mu(A_i) = 0$ for all i, the sum is 0. Since probability is non-negative, $\mu(\bigcup_{i=1}^{\infty} A_i) = 0$.

(c) We must show countable additivity. Let $A_1, A_2, \cdots \in \mathcal{F}$ be disjoint. By finite additivity, $\lambda(\bigcup_{i=1}^{N} A_i) = \sum_{i=1}^{N} \lambda(A_i)$. Since $\lambda \geq 0$, $\lambda(\bigcup_{i=1}^{\infty} A_i) \geq \lambda(\bigcup_{i=1}^{N} A_i) = \sum_{i=1}^{N} \lambda(A_i)$ for all N. Taking $N \to \infty$ gives $\lambda(\bigcup_{i=1}^{\infty} A_i) \geq \sum_{i=1}^{\infty} \lambda(A_i)$. The problem statement gives the sub-additive bound $\lambda(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \lambda(A_i)$. Combining these yields exact equality, proving λ is a measure.

[2] Let Ω be an uncountable set, and let $\mathcal{A}$ be an algebra on Ω.

(a) Prove that if $\mathcal{A}$ is closed under countable *disjoint* unions, then $\mathcal{A}$ is a σ-algebra.

(b) Let $\mathcal{C} = \{\{x\} : x \in \Omega\}$ be the collection of all singleton subsets of Ω. Describe the algebra $\mathcal{A}^* \equiv \mathcal{A}\langle\mathcal{C}\rangle =$ (smallest) algebra al generated by $\mathcal{C}$.

(c) Describe the σ-algebra $\sigma\langle\mathcal{C}\rangle$ generated by $\mathcal{C}$.

$(10 \times 3 = 30$ points)

(a) To show $\mathcal{A}$ is a σ-algebra, we must show it is closed under arbitrary countable unions. Let $A_1, A_2, \cdots \in \mathcal{A}$. We define a new sequence of sets: $B_1 = A_1$, and for $n \geq 2$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. Since $\mathcal{A}$ is an algebra, it is closed under finite unions and complements, so $B_n \in \mathcal{A}$ for all n. By construction, the sets B_n are pairwise disjoint, and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$. Because $\mathcal{A}$ is closed under countable disjoint unions, $\bigcup_{n=1}^{\infty} B_n \in \mathcal{A}$. Thus, $\mathcal{A}$ is a σ-algebra.

(b) Claim: $\mathcal{A}^* = \mathcal{F} \equiv \{E \subset \Omega : E$ is finite or E^c is finite$\}$ - the *finite-cofinite class*.

Since, $\{x\} \in \mathcal{F}$, all finite unions (all finite sets) and their complements (cofinite sets) must be in $\mathcal{A}^* = \mathcal{A}\langle\mathcal{C}\rangle$. So $\mathcal{F} \subseteq \mathcal{A}\langle\mathcal{C}\rangle \equiv \mathcal{A}^*$.

On the other hand, $\mathcal{C} \subseteq \mathcal{F}$ and the algebra $\mathcal{A}^* = \mathcal{A}\langle\mathcal{C}\rangle$ being the smallest algebra containing $\mathcal{C}$, we get $\mathcal{A}^* \subseteq \mathcal{F}$. This proves the claim.

(c) By a similar logic extended to countable operations, the σ-algebra generated by singletons is $\sigma\langle\mathcal{C}\rangle = \{E \subset \Omega : E$ is countable or E^c is countable$\}$.

[3] Let $F(x) = x^3$ for $x \in \mathbf{R}$, and let μ be the unique Lebesgue-Stieltjes measure on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ induced by F.

(a) For the standard Lebesgue measure m (where $F(x) = x$), we know $m(cA) = |c|m(A)$. Show that (using the Uniqueness Theorem) that the scaling relationship for $\mu(cA)$, $c > 0$ and $A \in \mathcal{B}(\mathbf{R})$ is:

$$\mu(cA) = c^3\mu(A)$$

(b) Show that it is rotation/reflection invariant, i.e.

$$\mu(-A) = \mu(A)$$

(c) Calculate the measure of the closed interval $[-2, -1]$, i.e., $\mu_F^*([-2, -1])$.

(d) Calculate the measure of the singleton set $\{3\}$, i.e., $\mu_F^*(\{3\})$.

$(10 \times 4 = 40$ points)

(a) Let's test the scaling on a generic right-closed interval $(a, b]$. By definition, $\mu((a, b]) = F(b) - F(a) = b^3 - a^3$. Now look at the scaled interval $c(a, b] = (ca, cb]$. $\mu(c(a, b]) = \mu((ca, cb]) = (cb)^3 - (ca)^3 = c^3b^3 - c^3a^3 = c^3(b^3 - a^3) = c^3\mu((a, b])$. Fix $c > 0$. Define two set functions on the measurable space $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$: $\nu_1(A) = \mu(cA)$ and $\nu_2(A) = c^3\mu(A)$. Because operations of scaling sets preserve the properties of σ-algebras, and μ is a measure, both ν_1 and ν_2 are valid measures on $\mathcal{B}(\mathbf{R})$. Let $\mathcal{C}$ be the semi-algebra of all right-closed intervals $(a, b]$, open rays (a, ∞), and $(-\infty, b]$. We just established that $\nu_1(A) = \nu_2(A)$ for all $A \in \mathcal{C}$. Because μ is finite on bounded intervals, both ν_1 and ν_2 are σ-finite on $\mathcal{C}$ (since $\mathbf{R} = \bigcup_{n \in \mathbf{Z}}(n, n + 1]$). Since $\mathcal{C}$ is a π-system that generates the Borel σ-algebra $\mathcal{B}(\mathbf{R})$, and the two σ-finite measures agree on $\mathcal{C}$, Caratheodory's Uniqueness Theorem dictates that they must agree on the entire generated σ-algebra. Thus, $\nu_1(A) = \nu_2(A)$, meaning $\mu(cA) = c^3\mu(A)$ for all $A \in \mathcal{B}(\mathbf{R})$.

(b) Just like in part (a) above, $\mu(-(a, b]) = \mu([-b, -a)) = (-a)^3 - (-b)^3 = b^3 - a^3 = \mu((a, b])$, using the fact that $F(x) = x^3$ is continuous. Proceeding as in part (a), we have the proof.

(c) For the interval was $[-2, -1]$, we evaluate $F(x)$ at the boundaries:

$$\mu_F([-2, -1]) = F(-1) - F(-2) = (-1)^3 - (-2)^3 = -1 - (-8) = 7$$

(d) Evaluation of $\mu_F(\{3\})$. For any singleton set $\{c\}$, the Lebesgue-Stieltjes measure evaluates the magnitude of the jump of $F(x)$ at c:

$$\mu_F(\{c\}) = F(c) - F(c^-)$$

Because $F(x) = x^3$ is strictly continuous everywhere on $\mathbb{R}$, there are no jumps in the function. Therefore, evaluating the singleton set at $x = 4$ yields:

$$\mu_F(\{4\}) = 4^3 - 4^3 = 0$$

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